RDX5 KIT SANDBOX A/B TEST

Objective: Determine the most practical long-term kit governance model for CAD → Composer → Partlink → ERP.

Test platform:

- Drifter: **7393 0800 45 – RDX5**
- CAD Identity: **7393 0800 45**
- OEM Identity: **BG 009 287 40**
- Existing Proreman dual-identity structure already separates CAD and OEM identities.
- Safe to ignore these documents [Kitify PDM...Setup.pdf | PDF], [KITIFY DUA...ECTURE.pdf | PDF]

Background

Meeting on 21 October 2026 identified that current PDM structure already supports:

- Proreman CAD identities (7300-series numbers)
- OEM identities stored as metadata
- Future ERP, Composer and Partlink mapping from metadata rather than filenames.

For a Complete Drifter Kit entry to exist in Partlink, we need:

- a Composer view
- geometry to explode
- geometry to isolate
- geometry to hotspot

The remaining question is:

Where should governed kits live?

Two candidate approaches will be evaluated.

OPTION A

Existing Proreman Approach + Improvements

CAD Structure

Keep current approach:

- Product sub-assemblies contain production hardware.
- Existing kit envelope/config methodology remains.
- No major CAD restructuring.

Example:

Front Head

├─ Seals

├─ Fittings

└─ Fasteners

Retain current config-based kit generation:

BEARING ONLY

FASTENERS ONLY

HYDR FITTINGS ONLY

But rename to:

Config name = Kit Part Number

Meta.PartNo = Kit Part Number

Meta.Description = Kit Description

Composer view-generation automation will now be able to create kit views with partno./description properties embedded in the source Sldwrks SLDASM file, no manual intervention required.

Pros:

- Removes manual Composer dummy assemblies.
- Removes manual per-view part number assignment.
- Kit metadata becomes reusable.

OPTION B

Create discrete governed kit assemblies

CAD Structure:

Front Head

└ **Front Head Kit**

└ Front Head Seal Kit.SLDASM

└ Front Head Fastener Kit.SLDASM

└ Front Head Hydr Kit.SLDASM

Impact Body

└ Impact Body **Kit**

└ Impact Body Seal Kit.SLDASM

└ Impact Body Fastener Kit.SLDASM

└ Impact Body Hydr Kit.SLDASM

Gear Housing

└ **Gear Housing Kit**

└ **Gear Housing** Seal Kit.SLDASM

└ **Gear Housing** Fastener Kit.SLDASM

└ **Gear Housing** Hydr Kit.SLDASM

Hardware ownership exists as real assemblies.

No envelope-only kit ownership.

Variant Handling

Engineering advice indicates:

If a Front Head design changes, a new 7372 Proreman identity is issued.

Front Head A.SLDASM

└ Front Head Seal Kit A.SLDASM

└ Front Head Fastener Kit A.SLDASM

└ Front Head Hydr Kit A.SLDASM

Front Head B

└ Front Head Seal Kit B.SLDASM

└ Front Head Fastener Kit B.SLDASM

└ Front Head Hydr Kit B.SLDASM

No cross-contamination between drifters.

Different drifters can remain on different Front Head generations.

Composer

Desired outcome:

Front Head

└ Front Head Kit

└ Front Head Seal Kit.SLDASM

└ Front Head Fastener Kit.SLDASM

└ Front Head Hydr Kit.SLDASM

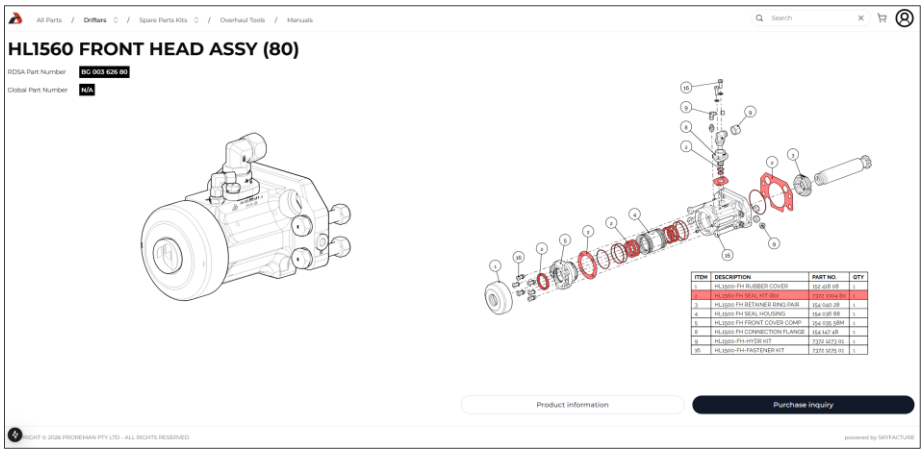
Appears naturally in Composer hierarchy allowing automated view generation.

No duplicate entities.

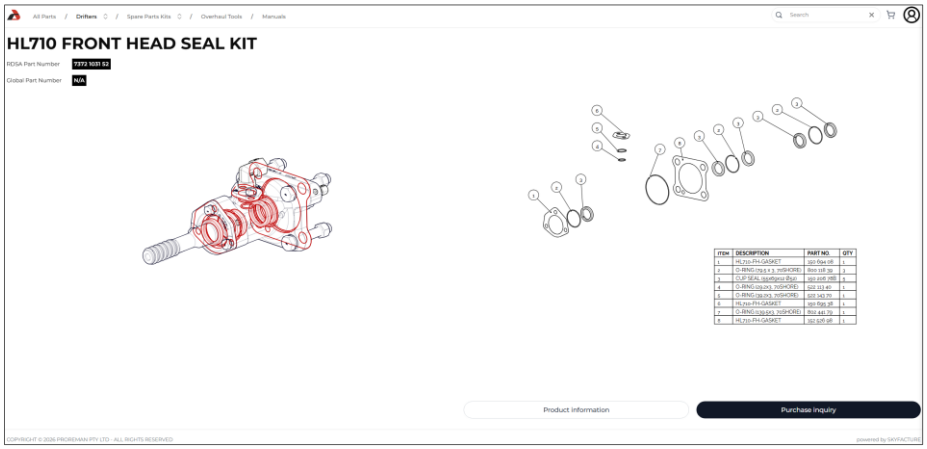
No envelope tricks.

Partlink

Kits can remain integrated and linked directly from exploded views



↓



Desired outcome:

Front Head Kit appears directly within:

- Front Head exploded view
- BOM hierarchy
- kit navigation

This remains faithful to Partlink's design intent, preserves the core Partlink supply chain and ecosystem, and is the premium solution for Partlink and marketing collateral such as CP brochures.

Complete Drifter Kits

A complete drifter kit should be evaluated independently from sub-assy kit ownership.

Complete drifter kits do not automatically require hardware ownership to live at drifter level.

The complete kit may exist either as:

1. **Named Kit Configuration / Envelope in the Main Drifter Assembly File**
   7393 0800 45.SLDASM with config: COMPLETE KIT

   or

2. **Separate Derived Envelope File**
   7393 XXXX XX COMPLETE KIT.SLDASM

Both approaches are valid provided the complete kit remains a derived representation.

Characteristics

- **Existing complete-kit workflow is preserved.**
- **Sub-assy kits remain the engineering source of truth.**
- **Complete kit becomes a presentation/catalog layer rather than a hardware ownership layer.**
- **Existing Composer workflows can continue with minimal disruption.**
- **Complete kit can carry its own metadata, part number and description.**

Rules

- **Front Head Kit, Impact Body Kit, Gear Housing Kit, etc. are the source of truth.**
- **Complete-kit representations reference those kits.**
- **Complete-kit representations do not own kit composition.**
- **Hardware changes are made only at sub-assy kit level.**
- **No hardware is created or maintained directly inside the complete-kit representation.**
- **Complete-kit content is derived from governed sub-assy kits.**

Pros

- **Preserves existing complete-kit workflows.**
- **Reduces Composer disruption (manual kit view creation reduced to one config vs multiple configs).**

- **Keeps ownership with saleable sub-assemblies.**
- **Allows complete-kit visualisation to remain largely unchanged.**
- **Provides a physical CAD representation of a complete drifter kit if required.**

Cons

- **Risk of ownership ambiguity if engineers begin editing kit content through the complete-kit representation rather than the governed sub-assy kits.**

Sandbox Evaluation

When evaluating Option B, determine:

Question 1

How should complete drifter kits be represented?

- **1 – Complete Kit config in main drifter file**
- **2 – Separate derived assembly envelope file**

Success Criteria

For either approach to be accepted:

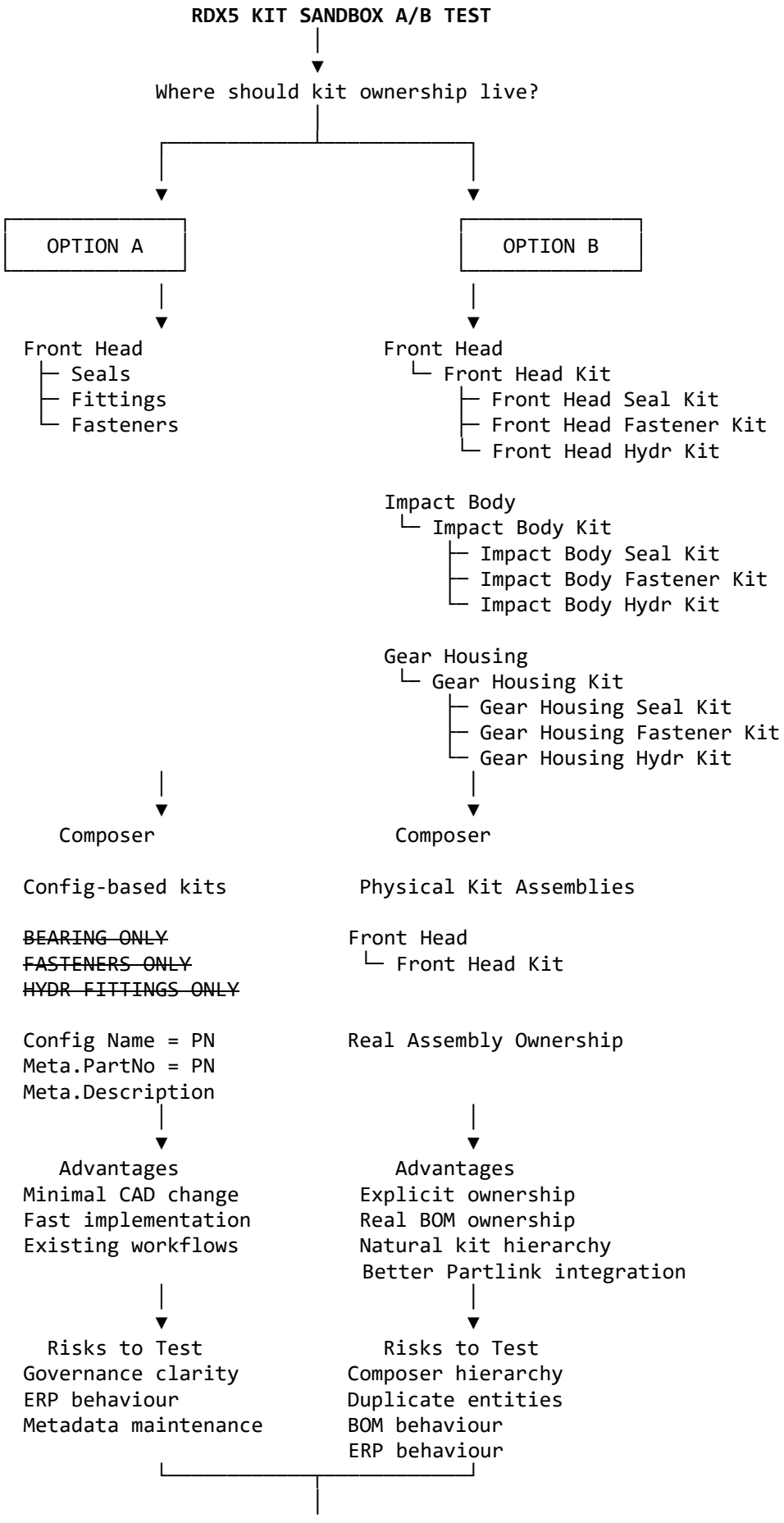
- No duplicate hardware.
- No duplicate Composer entities.
- Clean BOM generation.
- Simple engineering workflow.
- Supports saleable Front Heads/Gear Housings.
- Supports complete drifter kit generation.
- Supports future ERP integration.
- Requires minimal manual maintenance.

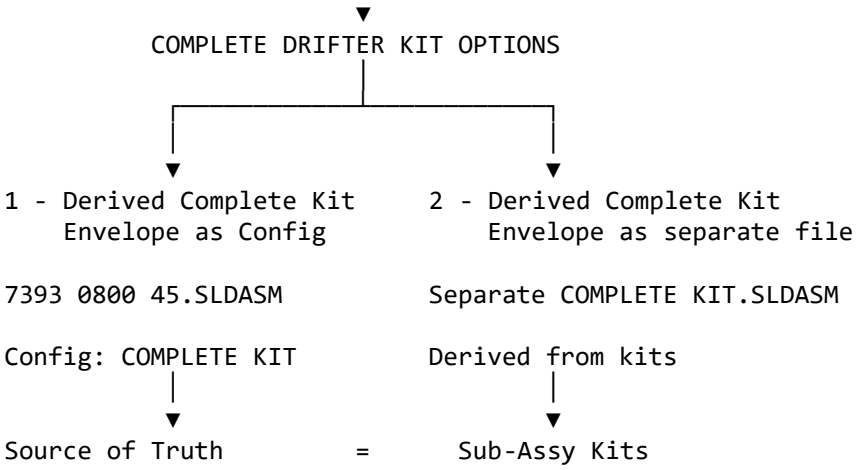
Recommendation

Do not debate architecture further until the RDX5 sandbox is tested.

Build both scenarios and evaluate:

- CAD effort
- Composer behaviour
- Partlink behaviour
- BOM clarity
- Maintenance overhead


RDX5 KIT SANDBOX A/B TEST
Where should kit ownership live?
OPTION A
OPTION B
Front Head
Seals
Fittings
Fasteners
Front Head
Front Head Kit
Front Head Seal Kit
Front Head Fastener Kit
Front Head Hydr Kit
Impact Body
Impact Body Kit
Impact Body Seal Kit
Impact Body Fastener Kit
Impact Body Hydr Kit
Gear Housing
Gear Housing Kit
Gear Housing Seal Kit
Gear Housing Fastener Kit
Gear Housing Hydr Kit
Composer
Composer
Config-based kits
BEARING ONLY
FASTENERS ONLY
HYDR FITTINGS ONLY
Config Name = PN
Meta.PartNo = PN
Meta.Description
Physical Kit Assemblies
Front Head
Front Head Kit
Real Assembly Ownership
Advantages
Minimal CAD change
Fast implementation
Existing workflows
Advantages
Explicit ownership
Real BOM ownership
Natural kit hierarchy
Better Partlink integration
Risks to Test
Governance clarity
ERP behaviour
Metadata maintenance
Risks to Test
Composer hierarchy
Duplicate entities
BOM behaviour
ERP behaviour



Complete Kit is always derived.
Complete Kit never owns hardware.
Hardware changes occur only at kit level.